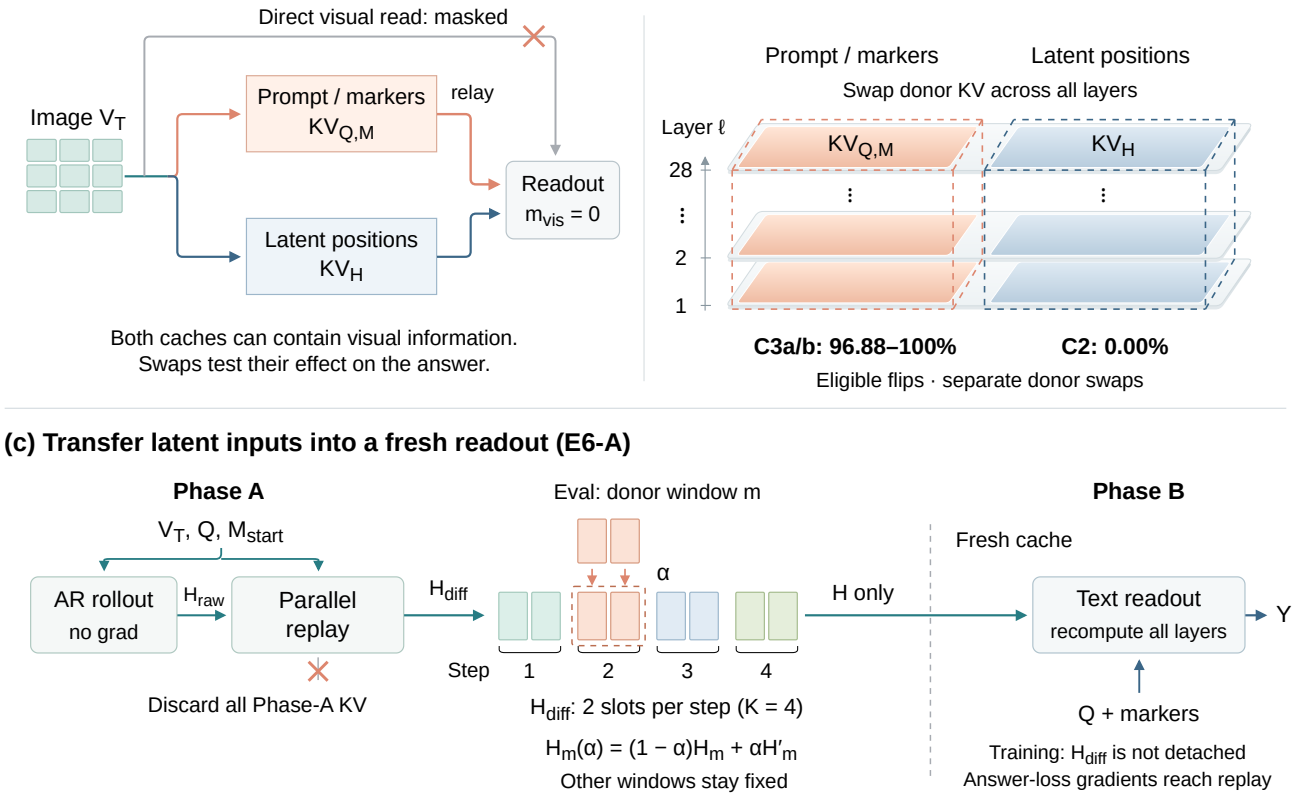
(a) Decode masking leaves cached routes
Direct visual read: masked
Image V_T
Prompt / markers
$KV_{Q,M}$
relay
Latent positions
KV_H
Readout
$m_{vis} = 0$
Both caches can contain visual information.
Swaps test their effect on the answer.
(b) Test the full cache depth (E6-C)
Prompt / markers
Latent positions
Swap donor KV across all layers
Layer ℓ
28
2
1
$KV_{Q,M}$
KV_H
C3a/b: 96.88–100%
C2: 0.00%
Eligible flips · separate donor swaps
(c) Transfer latent inputs into a fresh readout (E6-A)
Phase A
V_T, Q, M_{start}
AR rollout
no grad
H_{raw}
Parallel
replay
H_{diff}
Discard all Phase-A KV
Eval: donor window m
α
Step
1
2
3
4
H_{diff}: 2 slots per step (K = 4)
$H_m(\alpha) = (1 - \alpha)H_m + \alpha H'_m$
Other windows stay fixed
H only
Phase B
Fresh cache
Text readout
recompute all layers
Y
Q + markers
Training: H_{diff} is not detached
Answer-loss gradients reach replay